

Tagbly: Enhanced Multimedia

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ABSTRACT

Tagbly is an acronym for Text Audio Graphics AssemBLY. It is a method and system that synchronizes audio data with image data using an HTML control file. The HTML control file contains references to linked multimedia asset files and timestamps indicating when each of these multimedia assets will be displayed/played. Audiovisual content creators can incorporate interactive content, such as graphic elements (e.g., pictures, 360 images and 3D models), text-based annotations, clickable links and buttons, and zooming in/out of images. Viewers can interact with the interactive features while the multimedia content is being displayed/played. In the viewer's eye, Tagbly's audiovisual content looks like an "interactive video".

CCS CONCEPTS

• **Human-Centered computing**; • **Interactive systems and tools**; • **Applied computing**; • **Multi / mixed media creation**; • **Information systems**; • **Multimedia content creation**;

KEYWORDS

Enhanced multimedia, Interactive audiovisual presentations, Audiovisual synchronization

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1 INTRODUCTION

In the past 10 years, websites, blogs, video streaming service providers, and podcasts have become the main forms to disseminate information [5]. The popularization of digital media platforms (e.g., WordPress, YouTube and Apple Podcasts) has contributed to an increase in the production of video and audio content [8]. Currently, online content creators access multiple tools (e.g., audio and video editing software) and platforms (e.g., Facebook, Instagram and Twitter) to produce and share content [5]. However, creating online content may be overwhelming for creators with little to no knowledge of audio and video editing software. If a content creator desires to create content by synchronizing visual elements with audio, they will have no other option than creating a video. Furthermore, content creators promoting quizzes, sharing links or including any kind of interactive element on their content need to

redirect their audience to description sections or third-party service providers [5, 8].

We introduce Tagbly as a technology that synchronizes audio data with image data without the burden of creating a video. Content creators can create synchronized multimedia content using graphic elements (e.g., pictures, 360 images and 3D models), text-based annotations, video and audio files. As the assets in the multimedia content are not merged together, it allows modularity and makes it easy to implement future updates on the multimedia content. Tagbly links all these assets by keeping track of the time in which each asset will be displayed/played (i.e., timestamps). Furthermore, Tagbly enables content creators to incorporate interactive features such as clickable links and buttons, manipulation of 360 images and 3D models, and zooming in/out of images to multimedia content. Viewers can interact with these interactive features (e.g., manipulate images and click on links) while the multimedia content is playing. In the viewer's eye, Tagbly's multimedia content looks like an "interactive video".

2 BACKGROUND

Tagbly tackles three central issues in digital content creation: 1) significant effort required to create and edit multimedia content for the web; 2) creation of audiovisual content resulting exclusively on video files, thus requiring that content creators have video production skills; and 3) interactivity on multimedia content, currently explored by redirecting viewers to third-party service providers.

2.1 Creation of Multimedia Content for the Web

Producing and editing videos have become essential skills for almost anyone who desires to produce content for digital media platforms [5]. Nevertheless, non-technical users may perceive producing and editing videos as complex tasks [11]. Producing videos requires knowledge to operate video cameras and setup lighting. Editing video is even more complex, involving the learning of video editing software and editing skills. Furthermore, adding new elements and updates to a finished video can be as complex as creating a new video. The finalized video becomes a single file once it is created, thus requiring a re-rendering process to be updated [15]. That is, a content creator needs to keep the original video assets to update an existing video without losing its quality. Thus, producing and editing videos can be time consuming and overwhelming for non-technical users.

2.2 Limitations in Multimedia Content

The usage of video footage is not always a prerequisite for some kinds of online content. Some videos available on YouTube are narrated sequences of static images, stock videos and/or graphic elements [1, 4, 10]. Currently, an online content creator has the

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option to record an audio, and use video or image editing software (e.g., Adobe Premiere or Photoshop) or a slide presentation software (e.g., MS PowerPoint or Apple Keynote) to synchronize image/graphic data with audio data.

2.3 Interactivity on Multimedia Content

Online audiences have become interested in interacting with interactive elements in digital content [17]. A number of applications have been developed to enhance audiovisual content, incorporate interactive content over documents, and automatically generate audiovisual slideshows.

2.4 Enhanced Audiovisual Content

Apple developed an enhanced podcast application, allowing podcasters to link images, texts and chapters to podcast audio [3, 7, 12]. Yet a podcaster needs coding skills to embed these visual cues in the audio file.

The inclusion of images, links and geographic coordinates can help the audience to better experience the message conveyed by podcasters [16]. Recently, podcasters have used video platforms (e.g., YouTube) to stream podcasts - “videocast” [2]. Furthermore, some podcast applications have adopted machine learning techniques to create videos that synchronize visual elements with audio data [16]. Yet, the editing and updating process is still burdensome because the final result is a video file.

Online streaming services have offered interactive overlays so that users can include these elements on their videos. The interactive overlays contain clickable links, static and animated icons, and promoted ads [13, 14]. Recently, YouTube released the “video chapter” functionality which allows creators to add sections accompanied by previews to their videos [18]. Yet, the final format is a video file and the interactivity in video overlay functionality is limited.

2.4.1 Enhanced Multimedia Content. Applications have used computer vision algorithms and text-to-speech techniques to enhance documents and presentations [6, 9]. The computer vision feature matching algorithms and open database format have been used to incorporate interactive HTML 5 content to text-based documents and slide-based presentations without altering the source document [9]. Nevertheless, the user needs to register a new document whenever the augmented document is modified.

Another application automatically converts text to audiovisual slideshows by adopting text-to-speech techniques [6]. The application identifies “concrete” words in the text and automatically searches for images matching those concepts. The text is converted into voice-over narrations and synchronized with the images referring to the “concrete” words. In this case, users might not have the freedom to select the images that best represent the voice-overed text.

3 TAGBLY

Tagbly is an acronym for **T**ext **A**udio **G**raphics **A**ssem**B**ly. It is a method and a system implementing the creation of interactive multimedia content whose final result is not a video file. Although

video data can be included in the multimedia content, Tagbly preserves the assets in audio, image and video data by not merging these data into a single file.

We developed Tagbly’s synchronization method and system to work in any digital environment. Particularly, we have used the web platform (i.e., HTML and JavaScript) because it is free and familiar to both content creators and their audiences. The HTML digital control file has all the references to the asset multimedia files, as well as the timestamps indicating when each of these assets will be displayed/played. The digital control file can use any data structure, but currently it is implemented using HTML. The resulting digital file is not heavy, thus reducing the amount of data usage. Furthermore, Tagbly’s synchronized multimedia presentations can be easily indexed by search engines because it uses HTML.

Tagbly generates a master timeline synchronizing image data (i.e., still and moving images) with an audio timeline playback. The audio timeline contains timestamps indicating positions of time wherein each image data will be displayed. The master timeline establishes a time relationship between image data and the audio timeline to generate synchronized multimedia content. The synchronized multimedia content is an HTML digital control file containing the image and audio data assets to reproduce the synchronization. The HTML digital control file maintains the integrity of each image and audio data, thus making it easy to edit and update the synchronized multimedia presentation.

The creator’s interface contains tools to create, list and manipulate image and audio data. A content creator can upload and manipulate image data (e.g., resize, crop, color manipulating), and add text-based annotations (e.g., text and links) and interactive graphical elements (e.g., clickable buttons). The content creator can record, upload and manipulate audio data (e.g., arranging audio data in a sequential order, equalizing sounds, adding and/or removing parts of audio data, adding and/or removing sound effects). Once the content creator generates timestamps for each image data, the multimedia presentation playback will display each image data synchronized with the assigned point of time on the audio timeline.

Figure 1 illustrates the synchronized multimedia content frame-by-frame. The first frame shows an image assigned to be displayed when the audio starts to play. The second frame shows the same image zoomed in. The viewer has the option to zoom out or zoom in to see further details. A second image appears in sequence in the third frame, and then annotations supporting the audio content are displayed over the second image. Finally, a third image appears in the fifth frame and a few moments later external clickable links are shown over the third image.

Users can play the synchronized multimedia presentation on a web page media player application. The media player application parses the master timeline and assembles the pieces (e.g., audio and image data) comprising the multimedia presentation. The final playback file with audio and image data will play on the media player application in a similar format as video files. The synchronized multimedia presentation will look like a video containing interactive elements. While the synchronized multimedia presentation is being played, a user can click on links and buttons, zoom in and out images, rotate 360 pictures and 3D models.

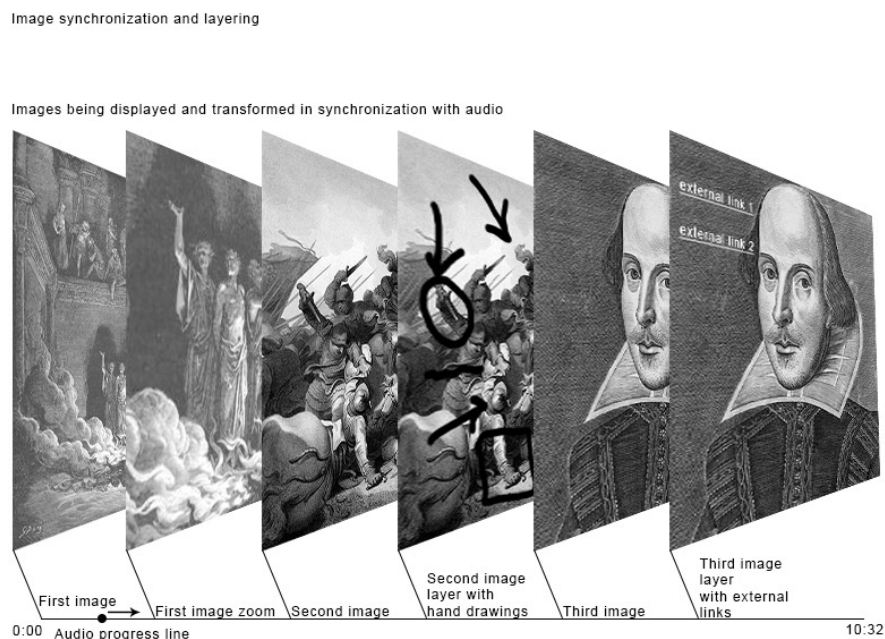


Figure 1: – Frame-by-frame illustration of Tagbly’s multimedia content synchronization.

3.1 Example Usage

3.1.1 Online Teaching. A tutor teaching online lessons wants to use an image to support the learning of their pupils while narrating the characteristics of such an image. The tutor uploads the supporting image and manipulates the image to be displayed during the online lessons. The tutor chooses to zoom in the image and display the details, thus supporting the narration describing the characteristics of the image. The tutor generates a timestamp and assigns this timestamp to the image. Consequently, the image will be displayed in a point of time coinciding with the audio timeline playback in a synchronized format. Rather than merging the image with the audio, Tagbly generates audiovisual content in a synchronized format establishing a timed connection between the image and audio data.

3.1.2 Social Media Interactions. A social media influencer wishes to share their expertise on makeup reviews and tutorials, and display the brands sponsoring their channel. The social media influencer takes pictures of their face wearing makeup and uploads them to Tagbly. The social media influencer adds links and quizzes to some of those pictures. The viewers can click on the links and be redirected to the social media influencer’s sponsors, and answer quizzes while the audiovisual content is being played.

3.1.3 Podcasting. A podcaster wants to show images of their birthday party while talking to their audience. The podcaster selects the images and chooses in what point of time on the audio timeline these images will be displayed. Additionally, the podcaster shares the clickable link to the bakery’s web page where they bought the birthday cake. The audience can zoom in the images to check the

details of the birthday cake and access the bakery’s web page while the podcast is playing.

3.1.4 Virtual Museums. A museum curator without any video editing experience wants to create an interactive online gallery for the museum’s collection. The museum curator records experts commenting on some collection items and takes pictures of these items. Then, the museum curator uploads image and audio data to Tagbly’s interface, and adds further information about these items (e.g., links, clickable buttons, quizzes). After the timestamps are assigned to the images and the master timeline is created, the museum curator generates an interactive audio tour for some items in the museum’s collection.

4 CONCLUSION

We introduced Tagbly, a method and a system to create interactive multimedia presentations and audiovisual content that does not result in video data. Tagbly’s method and system simplified the process of creating digital media by using web interfaces. Therefore, content creators are required to have basic knowledge of publishing digital media (e.g., blog posts) via web applications to use Tagbly.

We are currently working with the Beaty Biodiversity Museum at the University of British Columbia to develop interactive and multimedia content using Tagbly technology. We created a demonstration using Tagbly using parts of the content available on the Beaty Biodiversity Museum website. The full demonstration can be viewed at <https://tagbly.com/beaty-museum/00/00/08/>. Furthermore, a presentation introducing Tagbly’s technology is available on <https://tagbly.com>.

Tagbly allows better information indexing by not merging audio and image data assets together in one single file, as search engines access the structure of the presentations and return much more accurate results.

Tagbly is one step ahead to make interactive multimedia content more accessible and inclusive. It relies on well-established open technologies and makes it easier and cheaper to create content rich experiences.

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